Unit -1 (Introduction)

1. Introduction to Cloud Computing

1.1. Need for Cloud Computing

The need for cloud computing arises from the growing demand for efficient, scalable, and cost-effective ways to store, process, and manage data and applications.

Here are some of the need of Cloud Computing.

1. Cost Efficiency
2. Scalability and Flexibility
3. Accessibility and Mobility
4. Data Backup and Recovery
5. Speed and Performance
6. Focus on Core Business
7. Environmental Benefits
8. Security and Compliance

1.2. NIST Definition

“ Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

According to NIST, Cloud Computing has:

1. Five essential characteristics

-> On-demand self-service

Users can automatically access computing resources without human intervention.

-> Broad network access

Services are available over the network and accessed through standard devices (e.g., laptops, phones).

-> Resource pooling

Computing resources are shared among multiple users using a multi-tenant model.

-> Rapid elasticity

Resources can be quickly scaled up or down as needed.

-> Measured service

Usage is monitored, controlled, and billed based on Consumption.

1.3. Characteristics and benefits of Cloud Computing

=> Characteristics of Cloud Computing

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity (Scalability)
5. Measured service

=> Benefits of Cloud Computing

1. Cost Efficient
2. Scalability and Flexibility
3. Accessibility
4. Reliability and Backup
5. Performance Optimization
6. Security
7. Automatic Updates
8. Collaboration and Productivity

1.4. Applications of Cloud Computing

Cloud computing is widely used in various sectors and for different purposes. Some major applications include:

1. Data Storage and Backup
   - Cloud Storage services (like Google Drive, Dropbox, OneDrive, Amazon S3) allow users and organizations to store, access, and back up data online.
   - Ensures data availability and recovery in case of system failure.
2. Cloud Hosting and Web Applications
   - Websites, blogs, and applications can be hosted on cloud servers for better scalability and uptime.
   - Examples: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure.
3. Big Data Analytics
   - Cloud platforms offer tools for analyzing massive datasets in real time.
   - Used in business intelligence, market research, and decision-making.
   - Examples: Google BigQuery, AWS EMR, Azure Synapse.
4. Internet of Things (IoT)
   - Cloud provides centralized data storage and processing for IoT devices.
   - Enables smart homes, smart cities, and industrial automation.
   - Examples: AWS IoT Core, Azure IoT Hub.
5. Cloud Gaming
   - Games run on powerful cloud servers and are streamed to user's devices.
   - Reduces the need for high-end hardware.
   - Examples: NVIDIA GeForce Now, Xbox Cloud Gaming.
6. Artificial Intelligence and Machine Learning

- Cloud platforms offer AI/ML tools for developers to build and deploy models.
- Examples: Google AI platform, AWS SageMaker, Azure Machine Learning.

7. Business Applications

- Used in CRM (Customer Relationship Management), ERP (Enterprise Resource Planning), and HR management.
- Improves efficiency and collaboration.
- Examples: Salesforce, SAP cloud.

8. Education and E-Learning

- Cloud enables virtual classrooms, online learning, and data sharing among students and teachers.
- Examples: Google Classroom, Coursera, Microsoft Teams for Education.

9. Disaster Recovery

- Cloud-based disaster recovery solutions help restore data and systems after an outage or cyberattack.
- Reduces downtime and data loss.

2. Cloud Reference Model.

2.1. NIST Architecture

The NIST Cloud Computing Architecture is a conceptual framework developed by the National Institute of Standards and Technology (NIST) to define the key components and relationships in cloud computing.

It provides a standard model that helps understand how different parts of cloud computing (like consumers, providers, and services) interact with each other.

This conceptual architecture is built on five key actors who interact within cloud environments using defined service and deployment models, all based on a set of essential characteristics.

The five cloud actors:

1. Cloud Consumer: An individual or organization that purchases and uses cloud service from a cloud providers (e.g., using Gmail, Google Drive, AWS EC2).

2. Cloud Provider: The company that makes cloud services available to consumers (e.g., Amazon, Microsoft, Google).

3. Cloud Broker: Manages the use, performance, and delivery of cloud services, and negotiates relationships between providers and consumers.

4. Cloud Carrier: The intermediary that provides the connectivity and transport of cloud services from providers to consumers (e.g., Internet Service Providers).

5. Cloud Auditor: Conducts independent assessments of cloud services, such as security, performance, and compliance.

Cloud Service Models:

The NIST architecture identifies three main categories of cloud services:

1. Software as a Service (SaaS): The provider delivers a complete software application over the internet, with the consumer managing only limited configurations. Examples include Microsoft 365, Salesforce, Gmail, Google docs.

2. Platform as a Service (PaaS): The provider offers a platform for developing, running, and managing applications. Consumers control the applications, while the provider manages the underlying infrastructure and operating system. Examples include Heroku and AWS Elastic Beanstalk.

3. Infrastructure as a Service (IaaS): The provider offers virtualized computing resources like servers, storage, and networking. Consumers manage the operating systems and applications, while the provider handles the infrastructure. Examples include AWS EC2 and Google Compute Engine.

Cloud Deployment Models

The framework also outlines four standard ways to deploy cloud infrastructure:

1. Private Cloud: Exclusively used by a single organization, it can be managed internally or by a third party and located on-premise or off-premise.

2. Community Cloud: Provisioned for a specific group of consumers with shared concerns, such as a common mission or security requirements. (Cloud for universities or banks).

3. Public Cloud: Available for open use by the general public and owned and operated by a cloud provider. (Amazon Web Services (AWS), Microsoft Azure).

4. Hybrid Cloud: A combination of two or more distinct cloud infrastructures (private, community, or public) connected by technology. Example (AWS+ on-premise servers).

2.2 Design principles of Cloud Architecture

Here are the Design Principles of Cloud Architecture

1. Scalability

- The ability of a system to handle increasing loads by adding more resources (vertically or horizontally).
- Ensure the application can grow or shrink based on demand. Auto-scaling groups in AWS automatically add or remove servers as needed.

2. Elasticity

- The system automatically adjusts resources to match workload fluctuations.
- Optimize resource utilization and reduce costs. Scaling down servers during low traffic to save money.

3. Reliability and Fault Tolerance

- Ensuring the system continues to function correctly even when components fail.
- Avoid downtime and data loss.
- Methods
    - Redundancy (backup instances)

    - Failover systems
    - Data replication across regions

4. High Availability

- Designing systems to be continuously operational with minimal downtime. Maintain uninterrupted service.
- Deploying applications across multiple availability zones.

5. Security

- Protecting data, applications, and infrastructure from unauthorized access and attacks. Ensure confidentiality, integrity, and availability.
- Practices:
    - Encryption (in-transit and at-rest)
    - Identity and access management (IAM)
    - Network firewalls and monitoring

6. Performance Optimization

- Designing the system to deliver fast and consistent responses. Provide low-latency and high-throughput experiences.
- Techniques:
    - Content Delivery Networks (CDNs)
    - Load balancing
    - Caching mechanisms

7. Cost Efficiency

- Optimizing cloud spending by paying only for what you use.
- Achieve maximum performance at minimum cost.
- Techniques:
    - Right-sizing resources
    - Using spot or reserved instances
    - Monitoring usage with cost analyzers.

8. Automation

- Using scripts and tools to automate deployment, scaling, and management tasks.
- Reduce human errors and increase efficiency.

- Examples:
    - Infrastructure as Code (IaC) tools like Terraform, AWS CloudFormation
    - CI/CD pipelines for automated deployment

9. Loose Coupling

- Designing components so they are independent and interact through well-defined interfaces.
- Enhance flexibility and simplify updates.
- Examples: Using message queues or APIs between micro services.

10. Resilience and Recovery.

- Ensuring the system can recover quickly after failure.
- Minimize downtime and data loss.
- Practices:
    - Backup and disaster recovery plans.
    - Data replication across regions.

11. Observability and Monitoring

- Continuously tracking performance, logs, and metrics. Detect and response to issues quickly.
- Tools: CloudWatch, Prometheus, Grafana, etc.

12. Multi-Tenancy and Resource Pooling

- Serving multiple customers (tenants) using shared infrastructure.
- Efficient use of resources and cost savings.

2.3. Infrastructural Constraints

- Infrastructural constraints refer to the limitations and challenges related to the hardware, networking, storage, and data center infrastructure that supports cloud computing. These constraints can affect performance, availability, scalability, and cost-efficiency of cloud services.

1. Network Bandwidth and Latency

- Constraint: Limited internet bandwidth or high latency can slow down data transfer between users and cloud servers.
- Impact: Causes delays in accessing cloud applications or uploading/downloading large datasets.
- Example: Real-time video processing or IoT applications may suffer from lag due to latency issues.

2. Data Center Location and Availability

- Constraint: Cloud data centers are located in specific regions, which may be far from end users.
- Impact: Increases response time and may violate data residency laws or compliance requirements.
- Example: A business in Nepal may face latency if its data is stored in a data center located in Singapore or the US.

3. Power and Cooling Infrastructure

- Constraint: Data centers require constant power supply and efficient cooling systems.
- Impact: Power outages or cooling failures can lead to downtime and hardware damage.
- Example: In regions with unstable electricity supply, maintaining reliable cloud infrastructure is difficult.

4. Hardware Resource Limitations

- Constraint: Limited CPU, memory, or storage resources may affect performance during high demand.
- Impact: Overloaded servers can cause slow processing or service interruptions.
- Example: During peak hours, users may experience delays in cloud-hosted applications.

5. Security Infrastructure

- Constraint: Securing physical and virtual infrastructure requires strong firewalls, encryption, and intrusion detection systems.
- Impact: Weak infrastructure can lead to cyber-attacks, data breaches, and loss of trust.

- Example: Inadequate network segmentation can allow attackers to access sensitive data.

6. Dependency on Internet Connectivity
    - Constraint: Cloud services are entirely dependent on internet access.
    - Impact: Poor or unreliable connectivity disrupts access to cloud resources.
    - Example: In rural or remote areas with weak internet, users cannot effectively use cloud applications.
7. Scalability and Resource Allocation
    - Constraint: Although clouds are designed to scale, scaling physical infrastructure (like servers and storage) requires time and investment.
    - Impact: Limits the provider's ability to meet sudden spikes in user demand.
    - Example: A streaming service experiencing sudden viral traffic might face downtime if infrastructure isn't scaled fast enough.
8. Interoperability and Integration Issues.
    - Constraint: Different cloud providers use different infrastructure standards.
    - Impact Makes migration or integration between clouds difficult.
    - Example: Moving workloads from AWS to Azure may require reconfiguring network and storage settings.
9. Maintenance and Upgradation Challenges
    - Constraint: Hardware needs regular updates and replacements.
    - Impact: Maintenance downtime can temporarily affect service availability.
    - Example: During server upgrades, users might experience reduced performance.

3. Evolution of Cloud Computing

The phrase “Cloud Computing” was first introduced in the 1950s to describe internet-related services, and it evolved from distributed computing to the modern technology known as cloud computing. Cloud services include those provided by Amazon, Google, and Microsoft. Cloud computing allows users to access a wide range of services stored in the cloud or on the internet. Cloud computing services include computer resources, data storage, apps, servers, development tools, and networking protocols.

Early Foundations (1960s-1990s)

- 1960s: The foundational idea of computing as a utility, where resources could be shared among many users, emerged with the concept of distributed systems and the development of ARPANET, a precursor to the internet.
- 1990s: The rise of the internet made cloud computing possible. Virtualization technology began to develop, allowing multiple virtual machines to run on a single physical server, which was a crucial step toward more efficient resource use.

The birth of commercial cloud services (1990s – 2000s)

- 1999: Salesforce launched, pioneering the delivery of enterprise applications over the internet as a service (SaaS) and making one of the first commercial successes of cloud principles.
- 2002: Amazon began offering services through Amazon Web Services (AWS), laying the groundwork for its later public cloud platform.
- 2006: AWS officially launched Elastic Compute Cloud (EC2), which is widely considered the start of modern cloud computing, providing scalable, on-demand computing power that companies could rent.

Expansion and mainstream adoption (2000s –Present)

- 2000s: Other major tech companies, such as Microsoft with Windows Azure (now Microsoft Azure) and Google, entered the market, further driving adoption and competition.

- Present: Cloud computing has evolved into a complex ecosystem with distinct service models: IaaS, PaaS, and SaaS. Businesses now widely us hybrid and multi-cloud strategies to meet their diverse needs for flexibility and control. The technology continues to advance, with advancements in AI, machine learning, and the Internet of Things (IoT) heavily relying on cloud infrastructure for data storage and processing.

4. Cloud Service Model

=> Cloud computing offers various service models that cater to different needs, providing flexibility, scalability, and cost-effectiveness. These models define the level of control and management users have over the infrastructure, platform, and software. The primary service models are as follows:

4.1. Software as a Service (SaaS)

=> SaaS delivers fully functional software applications over the internet. Users access these applications via a web browser without managing infrastructure or updates.

Example: Google Workspace (Docs, Sheets) or Salesforce.

Advantages:

- No setup or maintenance required.
- Accessible from any device with an internet connection.
- Subscription-based pricing.

Disadvantages:

- Limited customization options.
- Dependence on the provider for security and availability.

4.2. Platform as a Service (PaaS)

=> PaaS offers a platform for developers to build, deploy, and manage applications without worrying about the underlying infrastructure. The provider manages servers, storage, and runtime environments.

Example: Developing web apps using Google App Engine or AWS Elastic Beanstalk.

Advantages:

- Simplifies application development with pre-configured environments.
- Built-in scalability and developer tools.
- Reduces infrastructure complexity.

Disadvantages:

- Limited customization and flexibility.
- Vendor lock-in risks.

4.3. Infrastructure as a Service (IaaS)

=> IaaS provides virtualized computing resources such as servers, storage, and networking. Users manage the operating system, applications, and data, while the provider handles the physical infrastructure.

Example: Hosting a website using AWS EC2 or Google Compute Engine.

Advantages:

- High scalability and flexibility.
- Pay-as-you-go pricing reduces upfront costs.
- Suitable for businesses needing full control over infrastructure.

Disadvantages:

- Requires expertise in system administration.
- Users are responsible for securing their data and applications.

Feature	IaaS	PaaS	SaaS
User Control	High	Medium	Low
Managed by Provider	Hardware	Hardware + OS + Middleware	Everything
Target Users	System Admins	Developers	End Users
Examples	AWS EC2, Azure VM	Google App Engine	Gmail, Salesforce

Extended Models: Some of the emerging models are as follows:

i. FaaS (Function as a Service)

ii. XaaS (Anything as a Service)

5. Cloud Deployment Model

=> Cloud deployment models describe how cloud services are made available to users – who owns the infrastructure, how it is accessed.

There are four main deployment models in cloud computing: Public, Private, Hybrid, and Community Cloud.

5.1. Public Cloud

=> A public cloud is owned and managed by third-party cloud providers and delivers services over the internet to multiple users.

Key Features:

- Shared infrastructure among multiple customers (multi-tenant environment)
- High scalability and cost-effectiveness
- Users pay for what they use (pay-as-you-go model)

Advantages:

- Low cost (no hardware investment)
- Easy scalability and flexibility
- Maintenance handled by provider

Disadvantages:

- Less control over security and customization
- Data privacy concerns

Examples:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Use Cases:

- Web-based email, storage, and collaboration tools
- Application hosting and testing

2. Private Cloud

=> A private cloud is used exclusively by a single organization. It can be hosted on-premises or by a third-party provider, but the resources are not shared with others.

Key Features:

- Dedicated infrastructure for one organization
- Greater control, security, and customization

Advantages:

- Enhanced security and privacy
- Customizable to meet business requirements
- Better compliance with data regulations

Disadvantages:

- High cost (requires hardware and maintenance)
- Limited scalability compared to public cloud.

Examples:

- VMware vSphere
- Microsoft Azure Stack
- OpenStack

Use Cases:

- Banking, government, and healthcare organizations needing data security

3. Hybrid Cloud

=> A hybrid cloud combines public and private clouds, allowing data and applications to be shared between them.

Key Features:

- Flexible deployment across multiple environments
- Sensitive data can stay on private cloud; general workloads on public cloud.

Advantages:

- Optimized cost and performance
- Greater flexibility and scalability
- Business continuity and disaster recovery

Disadvantages:

- Complex integration and management
- Security challenges between environments

Examples:

- AWS Outposts
- Microsoft Azure Hybrid
- Google Anthos

Use Cases:

- Organizations balancing sensitive and non-sensitive workloads
- Backup and disaster recovery solutions

4. Community Cloud

=> A community cloud is shared by several organizations with common goals, policies, or compliance requirements.

Key Features:

- Shared infrastructure for a specific community (e.g., universities, healthcare groups)
- Managed internally or by a third party

Advantages:

- Cost shared among organizations
- Collaborative platform for similar needs
- Improved security compared to public cloud

Disadvantages:

- Higher cost than public cloud
- Limited scalability

Examples:

- Government community clouds
- Healthcare or education cloud systems

Use Cases:

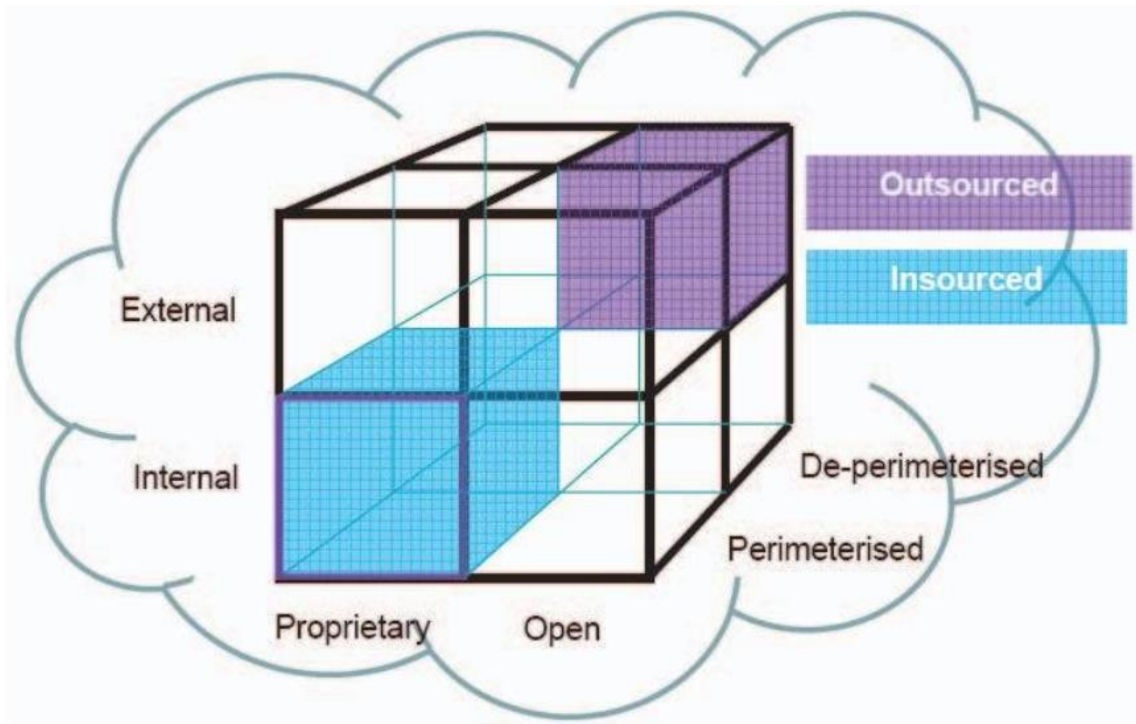
- Research collaborations, government departments, healthcare systems.

Feature	Public Cloud	Private Cloud	Hybrid Cloud	Community Cloud
Ownership	Service provider	Single organization	Mixed (both)	Group of organizations
Security	Moderate	High	High (configurable)	Medium–High
Cost	Low	High	Moderate	Shared
Scalability	High	Limited	High	Moderate
Control	Minimal	Full	Partial	Shared

Jericho Cloud Cube Model

- ⇨ The Jericho Cloud Cube Model (aka the Cloud Cube Model, CCM) is a simple but powerful framework for classifying cloud formations along four independent dimensions so you can reason about where security controls must sit and which cloud patterns fit your needs.
- ⇨ The Jericho Forum created the Cloud Cube Model to show that "cloud" deployments can be classified along four independent axes

(so you can pick a secure formation that matches your risk, compliance and control needs).



The four dimensions

Each dimension is a binary axis. Combine them and you get different cube positions (i.e., cloud formations):

1. Internal/External -- Is the cloud inside your organization’s boundary (private/internal) or outside it (public/external)?

2. Proprietary/Open – Is the service proprietary (single vendor, closed) or built on open standards and interoperable components?

3. Perimeterised/De-perimeterised – Does the solution rely on a traditional security perimeter (firewalls, network boundary) or assume a de-perimeterised architecture where security is applied to data and entities rather than the network? (This ties into the Jericho Forum’s broader work on de-perimeterisation).

4. Insourced/ Outsourced – Is the service delivered and managed in-house (insourced) or by an external provider (outsourced)?

How to read/use the model

- Think of each axis as a yes/no choice. Picking values on the four axes places your workload into one corner of the cube – that tells you expected control, sharing, and threat model.
- Use it when assessing: regulatory constraints, data residency, trust in vendors, need for interoperability, and whether you can rely on a network perimeter for protection.

Quick examples

- Public SaaS: External/Proprietary/De-perimeterised/Outsourced – low control, higher vendor trust required.
- Private, in-house IaaS: Internal/Proprietary or Open/Perimeterised/Insourced – higher control, good for regulated data.
- Community cloud for a consortium: External/Open/Perimeterised or De-perimeterised/ Outsourced
  - Collaborative but with governance/standards.
    (These are illustrative, real workloads can mix axes).

Why it's useful

- Forces you to state tradeoffs explicitly (control vs cost vs interoperability).
- Helps pick the right protection model (e.g., if de-perimeterised, focus on encryption/identity/data controls rather than network firewalls).
- Useful for vendor selection and compliance checks (where "external+outsourced" may need stronger contractual safeguards).

7. Challenges and Ethical Issues

=> Challenges in Cloud Computing

1. Security and Privacy

- Major concern since data is stored on third-party servers.
- Risks include data breaches, unauthorized access, data loss, and insider threats.

- Encryption, access control, and compliance standards are critical to mitigate risks.

2. Data Ownership and Control
   - Once data is uploaded, the ownership and control can become unclear.
   - Users may lose control over how their data is managed or shared by the provider.
3. Downtime and Reliability
   - Cloud services depend on internet connectivity and provider uptime.
   - Service outages or network failures can disrupt business operations.
4. Vendor Lock-In
   - Difficult to switch providers due to incompatible APIs, platforms, or data formats.
   - Reduces flexibility and may increase long-term costs.
5. Compliance and Legal Issues
   - Cloud data may be stored in different countries with varying data protection laws.
   - Meeting compliance requirements (like GDPR, HIPAA, etc.) becomes complex.
6. Data Transfer and Bandwidth
   - Uploading or accessing large volumes of data can be slow and costly.
   - Latency can affect performance for time-sensitive applications.
7. Limited Transparency
   - Users often have limited visibility into how providers manage data and security.
   - Lack of transparency can reduce trust.
8. Performance Issues
   - Shared cloud resources may cause fluctuating performance depending on demand.

Ethical Issues in Cloud Computing

1. Data Privacy and Confidentiality
   - Users must trust providers not to misuse or sell personal data.
   - Ethical issue arises when cloud vendors collect or analyze data without consent.
2. Data Ownership and Intellectual Property
   - Who owns the data stored in the cloud – the user or the service provider?
   - Providers must not claim ownership or exploit user data for profit.
3. Transparency and Trust
   - Providers should clearly disclose how data is stored, processed, and shared.
   - Ethical obligation to inform customers about security breaches or policy changes.
4. Digital Divide
   - Unequal access to high-speed internet and cloud resource creates inequality.
   - Raises ethical concerns about fairness and accessibility.
5. Environmental Impact
   - Large data centers consume huge amounts of energy and resources.
   - Ethically, providers should invest in green computing and renewable energy.
6. Surveillance and Government Access
   - Governments may request access to user data without consent.
   - Raises ethical questions about privacy vs. national security.
7. Misuse of Cloud Resources
   - Cloud platforms can be exploited for cyber attacks, piracy, or hosting illegal content.
   - Providers have an ethical duty to monitor and prevent misuse responsibly.

7. Challenges and Ethical Issues

=> Cloud computing offers flexibility and cost efficiency but faces several challenges and ethical concerns. Major challenges include:

- Data security and privacy: As information stored on third-party servers is vulnerable to breaches and unauthorized access.
- Data ownership and control: Are often unclear, and vendor lock-in makes it difficult to switch providers.
- Other challenges include service downtime, compliance with international laws, limited transparency, and performance issues due to shared resources.

Ethical Issues arise when providers misuse or share user data without consent, raising questions about privacy, trust, and data ownership. The environmental impact of large data centers, digital divide due to unequal access, and government surveillance also pose ethical dilemmas. Therefore, ensuring transparency, fairness, and responsible data handling is vital for trustworthy cloud computing.

THE END